

Simulation-Based Analysis of a Liquid-Cooled Data Center with Waste Heat Reuse

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Abstract

Driven by rapidly growing environmental footprint, improving the energy efficiency of data centers (DC) has become increasingly urgent. Heat pump (HP) integration is a promising way to enable the reuse of low-grade waste heat from liquid-cooled DCs. Increasing the liquid cooling system (LCS) supply temperature generally improves HP efficiency. However, higher supply temperature can increase IT power consumption and shift more heat to the server room air, potentially raising the overall DC power consumption. These coupled interactions between the DC and the HP are often overlooked. This work addresses this gap by developing a holistic model of a direct liquid-cooled DC with HP-assisted waste heat reuse that captures the critical dynamics of the LCS and air cooling system (ACS), energy performance of HPs, temperature dependence of IT leakage power, and heat transfer from IT equipment to the server room. The model is applied to the HAWK supercomputer at the High-Performance Computing Center Stuttgart, Germany, to study the impact of heat reuse on DC power consumption and how the LCS supply temperature affects the coupled operation between the DC and the HP-assisted heat reuse system. Results show that heat reuse can reduce DC power consumption when the supply temperature is selected appropriately. However, even under full heat reuse, raising the supply temperature beyond an appropriate range can increase DC power because reduced LCS power can be outweighed by higher IT and ACS power. From a coupled-system perspective, although operating at the maximum permitted supply temperature maximizes heat pump efficiency, it can increase the combined power consumption relative to the optimal operating point, with increases of up to 9.5% observed. The findings highlight the need for dynamically optimized control of the supply temperature to achieve robust benefits and demonstrate the importance of holistic modeling for evaluating liquid-cooled DCs with heat reuse.